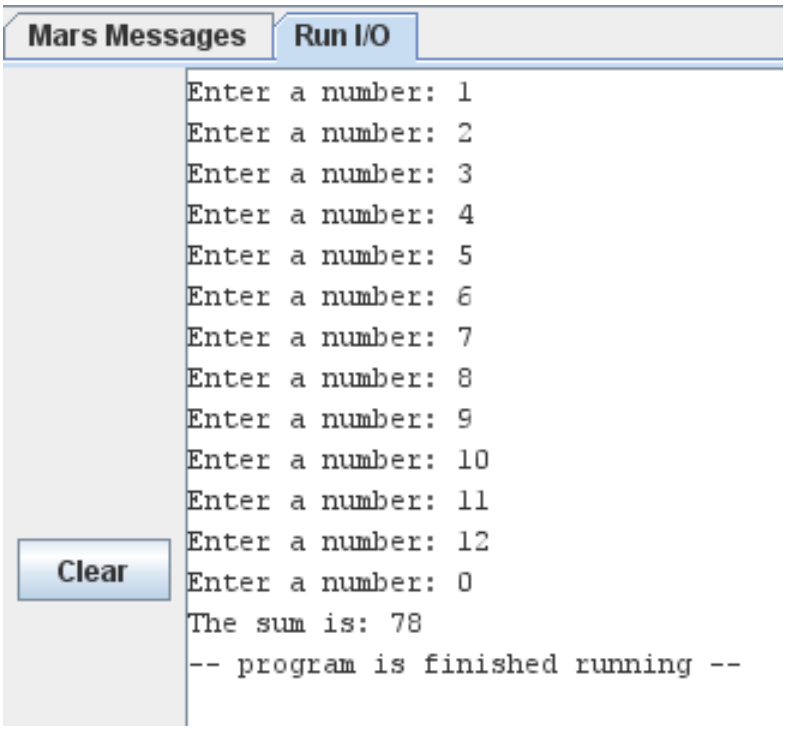
Lab4-1 CO Report

1. Austin Sternberg
2. lab4-1.asm
3. The program is designed to continuously add integers until the user inputs a zero. Lines 20-23 initialize the sum value ($t0) to zero to prime it for the incoming values. Values are read through a loop that exists between lines 24-42. The first part of this loop, lines 26-28, prompt the user to input an integer, and the next part, lines 31-33, read that value into register $t1. From there, line 36 checks if that value is zero. If it is zero, the program jumps to the end section and breaks out of the loop, but if it is not zero the loop will continue. Lines 39-42 of the loop add the integer to the current sum (stored in $t0), then jump back to line 24 to restart the loop. One we get to the print_sum section, line 44-end, the code prints out the sum value in register $t0 then ends the program.
4. 

5. Honestly this part of this assignment was pretty straightforward. Nothing required reading ahead or researching how functions worked and how to implement them. Everything necessary to make this one work had already been used in the previous 3 programming assignments. The only piece that took a bit of effort was combining it all together for this project, but even that was not all that difficult. I did learn that registers can hold a considerable sized value before maxing out.